

RB-LS: Let  $T$  be any CSS to  $\theta$  and  $\delta$  any u.e. of  $\psi(\theta)$ .

Para Int  
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Then  $\phi(T) = E(\delta|T)$  is the UMVUE of  $\psi(\theta)$

Strategy I (Direct conditioning) Find  $T = \text{C.S.S.}$  and  $\delta$ . Then compute  $E(\delta|T)$

Ex  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bern}(\theta)$  UMVUE of  $\psi(\theta) = \theta^r$   $r \leq n, r \in \mathbb{N}$   
$$\text{UMVUE} = \frac{\binom{T}{r}}{\binom{n}{r}} \mathbb{1}_{(r \leq T)} \quad \text{or} \quad \frac{\binom{n-r}{T-r}}{\binom{n}{r}} \mathbb{1}_{(r \leq T)} \quad (\text{same})$$

Strategy II

1. Find CSS  $T$
2. Find a function  $\delta(T)$  that is an u.e. of  $\psi(\theta)$   
i.e. solve  $E_{\theta}[\delta(T)] = \psi(\theta) \quad \forall \theta$   
works b/c.  $E(\delta(T)|T) = \delta(T)$

Ex 1.  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ . Find UMVUE of  $\psi(\lambda) = e^{-\lambda}$   
A CSS for  $\lambda$  is  $T = \sum_{i=1}^n X_i$  (why?)

Suppose  $\delta(T)$  u.e. of  $e^{-\lambda}$   
 $\sim \text{Poi}(n\lambda)$

i.e.  $E(\delta(T)) = e^{-\lambda}$

$$\Rightarrow \sum_{t=0}^{\infty} \delta(t) \frac{e^{-n\lambda} (n\lambda)^t}{t!} = e^{-\lambda} \quad \forall \lambda > 0$$

$$\Rightarrow \sum_{t=0}^{\infty} \delta(t) \frac{n^t \lambda^t}{t!} = e^{(n-1)\lambda} = \sum_{t=0}^{\infty} \frac{(n-1)^t \lambda^t}{t!} \quad \forall \lambda > 0$$

$$\Rightarrow \delta(t) = \left(\frac{n-1}{n}\right)^t \quad \forall t = 0, 1, 2, \dots$$

$\therefore \delta(T) = \left(1 - \frac{1}{n}\right)^T$  is the UMVUE.

Ex.  $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ . Find UMVUE of  $\mu, \sigma^2$ ;  $\xi_p = p^{\text{th}}$  quantile of  $N(\mu, \sigma^2)$

Find CSS for  $(\mu, \sigma^2) \Rightarrow T = (\bar{X}, S^2)$

$\therefore \bar{X}$  u.e. of  $\mu$  that is fn. of  $T = (\bar{X}, S^2)$  so,  $\bar{X}$  UMVUE

Similarly  $S^2$  UMVUE of  $\sigma^2$

$$p = P[X_1 \leq \xi_p] = P\left[\frac{X_1 - \mu}{\sigma} \leq \frac{\xi_p - \mu}{\sigma}\right] \\ = \Phi\left(\frac{\xi_p - \mu}{\sigma}\right)$$

$$\Phi^{-1}(p) = \frac{\xi_p - \mu}{\sigma} \Rightarrow \xi_p = \mu + \sigma \underbrace{\Phi^{-1}(p)}_{\text{known}}$$

u.e. of  $\mu = \bar{X}$

u.e. of  $\sigma$

function of  $T \rightarrow = \frac{\sqrt{n-1}}{C(n)} S$

$$Y = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$E Y^{1/2} = \frac{\sqrt{n-1}}{\sigma} E S$$

$$\stackrel{||}{C(n)}$$

some constant

$\therefore$  UMVUE of  $\xi_p$  is

$$\bar{X} + \frac{\sqrt{n-1}}{C(n)} \Phi^{-1}(p) \sqrt{S^2}$$

Ex.  $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$ ,  $\theta > 0$  Find UMVUE of  $\theta^r$   $r > 0$   
CSS for  $\theta$  is  $T(\underline{X}) = X_{(n)}$ . Suppose  $\delta(T) =$  u.e. of  $\psi(\theta)$

Need to solve  $E_{\theta}(\delta(T)) = \theta^r \forall \theta$

$$\int_0^\theta \delta(t) \frac{n}{\theta^n} t^{n-1} dt = \theta^r \quad \forall \theta > 0$$

$$\Rightarrow \int_0^\theta \delta(t) t^{n-1} dt = \frac{\theta^{n+r}}{n} = \int_0^\theta \frac{n+r}{n} t^{n+r-1} dt$$

$$\Rightarrow \int_0^\theta \underbrace{\left[ \delta(t) - \frac{n+r}{n} t^r \right]}_{f(t)} t^{n-1} dt = 0 \quad \forall \theta > 0$$

Using Lebesgue integration theory

$$\delta(t) = \frac{n+r}{n} t^r, \quad t > 0$$

$\frac{n+r}{n} X_{(n)}^r$  is UMVUE of  $\theta^r$

$r=1$   $\frac{n+1}{n}$  is UMVUE of  $\theta$

$$E[2X_1 | X_{(n)}] = \frac{n+1}{n} X_{(n)}$$

b/c  $2X_1$  u.e. of  $\theta$  and  $\frac{n+1}{n} X_{(n)}$  UMVUE of  $\theta$

A "iff" condition for u.e.  $\delta$  to be UMVUE of  $\psi(\theta)$

Define  $\mathcal{U} = \{ \delta : E_\theta \delta = \psi(\theta), E_\theta \delta^2 < \infty \quad \forall \theta \}$

= class of u.e. of  $\psi(\theta)$

$\mathcal{U}_0 = \{ e : E_\theta e = 0, E_\theta e^2 < \infty \quad \forall \theta \}$

= class of u.e. of 0

Let  $\delta \in \mathcal{U}$ .  $\delta$  is UMVUE iff  $\text{cov}(\delta, e) = E(\delta e) = 0$

i.e.  $\delta$  must be uncorrelated with all u.e. of 0.  $\forall e \in \mathcal{U}_0$

Pf:  $(\Rightarrow)$  Given  $\delta = \text{UMVUE}$

$$\underbrace{\delta' = \delta + \lambda e}_{\in \mathcal{U}} \text{ where } e \in \mathcal{U}_0, \lambda \in \mathbb{R}$$

$$\delta' \in \mathcal{U} \quad \text{Var } \delta' = \cancel{V(\delta)} + \lambda^2 V(e) + 2\lambda \text{Cov}(\delta, e) \geq V(\delta)$$

$$\lambda^2 V(e) + 2\lambda \text{Cov}(\delta, e) \geq 0 \quad \forall \lambda \in \mathbb{R}$$

$$\text{Take } \lambda = -\frac{\text{Cov}(\delta, e)}{V(e)} \text{ using } \textcircled{*}$$

$$\Rightarrow -\frac{\text{Cov}^2(\delta, e)}{V(e)} = 0 \Rightarrow \text{Cov}(\delta, e) = 0 \quad \forall e \in \mathcal{U}_0$$

$(\Leftarrow)$  Given  $\delta \in \mathcal{U}$ ,  $\text{Cov}(\delta, e) = 0 \quad \forall e \in \mathcal{U}_0$

Take  $\delta' \in \mathcal{U}$  def.  $e^* = \delta' - \delta \in \mathcal{U}_0$

$$\text{Cov}(\delta, e^*) = 0$$

$$\text{Cov}(\delta, \delta' - \delta) = 0$$

$$\text{Cov}(\delta', \delta) = \text{Var } \delta$$

$$\underbrace{\text{Cov}^2(\delta', \delta)}_{V(\delta)^2} \leq V(\delta') V(\delta)$$

$$\Rightarrow V(\delta) \leq V(\delta') \quad \square$$

Uniqueness of UMVUE

Suppose  $\delta_1, \delta_2$  UMVUE of  $\psi(\theta)$

$$V(\delta_1) = V(\delta_2) \quad \text{Cov}(\delta_1, \delta_1 - \delta_2) = 0 \quad \left[ \because \delta_1 = \text{UMVUE and } \delta_1 - \delta_2 \in \mathcal{U}_0 \right]$$

$$\Leftrightarrow \text{Cov}(\delta_1, \delta_2) = V(\delta_1) = V(\delta_2)$$

$$V(\delta_1 - \delta_2) = V(\delta_1) + V(\delta_2) - 2\text{Cov}(\delta_1, \delta_2) = 0$$

Applications Suppose  $T$  any stat,  $\phi(T)$  any real f.n. with finite 2<sup>nd</sup> moments is u.e. of  $E_\theta(\phi(T)) := \psi(\theta)$

g. When  $\phi(T)$  is UMVUE

Need.  $\text{Cov}(\phi(T), e) = 0 \quad \forall e \in \mathcal{U}_0 \rightarrow$  the class is difficult to capture

If we require  $T$  to be complete

$$E[g(T)] = 0 \Rightarrow g(T) = 0$$

$f(x|\theta) \rightarrow T(X) \sim f_T(t|\theta)$  the only u.e. of 0  
(i.e. based on  $T(X)$  is 0 itself)  
 $\Rightarrow \text{Cov}(\phi(T), 0) = 0 \Rightarrow \phi(T)$  UMVUE of  $\psi(\theta)$

(2) Suppose  $\delta_i$  is UMVUE of  $\psi_i(\theta)$  for  $i=1, \dots, k$   
Then  $\sum_{i=1}^k a_i \delta_i$  is the UMVUE of  $\sum_{i=1}^k a_i \psi_i(\theta)$

Given  $\text{Cov}(\delta_i, e) = 0 \quad \forall e \in \mathcal{U}_0$

$$\Rightarrow \text{Cov}\left(\underbrace{\sum_{i=1}^k a_i \delta_i}_{\text{u.e. of } \sum_{i=1}^k a_i \psi_i(\theta)}, e\right) = 0$$

Note: UMVUE may exist even if c.s.s. does not exist

Ex. Suppose  $X$  has pmf  $f(x|\theta) = \begin{cases} \theta & \text{if } x = -1 \\ (1-\theta)^2 \theta^x & \text{if } x = 0, 1, 2, \dots \\ 0 & \text{o/w} \end{cases} \quad 0 < \theta < 1$

Find UMVUE of  $\psi(\theta) = (1-\theta)^2$

$X$  is sufficient

$E_\theta[X] = 0 \quad \forall \theta$  but  $P_\theta[X=0] \neq 1 \Rightarrow X$  is not complete

$\delta(X) = 1(X=0)$  u.e. of  $\psi(\theta)$

$\text{Cov}(\delta(X), c) = 0 \quad \forall c \in \mathcal{U}_0$

$\Leftrightarrow E(\delta(X)c) = 0$

$\therefore \mathcal{U}_0 = \{cX : c \in \mathbb{R}\}$

take any  $cX \in \mathcal{U}_0$

$E[\delta(X)cX]$

$= c E[X 1_{\{X=0\}}] = 0 \quad \forall c \in \mathbb{R}$

$\Rightarrow \delta(X)$  is the UMVUE of  $\psi(\theta)$

But  $\delta(X)$  is not a function of c.s.s. of  $\theta$

Suppose:

$h(X) \in \mathcal{U}_0$

$E_\theta h(X) = 0 \quad \forall \theta$

$\theta h(-1) + \sum_{x=0}^{\infty} h(x) (1-\theta)^2 \theta^x = 0$

$h(0) + \sum_{x=0}^{\infty} (h(x+1) - 2h(x) + h(x-1)) \theta^{x+1} = 0$

$h(x+1) = 2h(x) - h(x-1) \quad \forall \theta$   
 $h(0) = 0$

$\Rightarrow h(X) = cX$  for  $c = -h(-1) \in \mathbb{R}$

Result: If  $T$  is a complete